

## 2 (short)

A small open reservoir containing water is shown in Fig. 2. The retaining structure on the left hand side of the reservoir includes a flap sluice gate that can be used to release water from the reservoir. The flap is a plate 1 m wide and 3 m high that is supported from above by a horizontal hinge at A. A horizontal ram is used to control the opening of the flap. If the water is 6 m deep and the flap is closed, what is the magnitude of the force  $R$  on the ram?

[10]

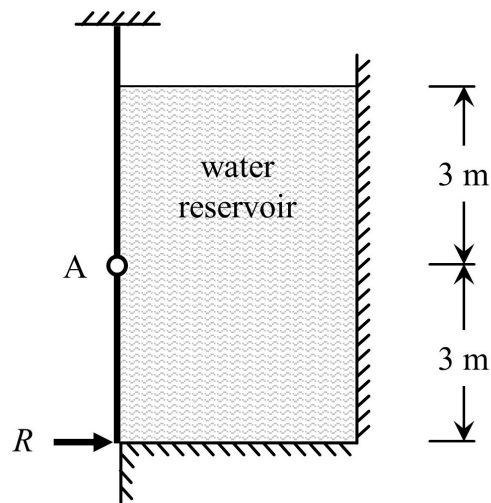


Fig. 2